

Dating wide-field images of the night sky with planetary ephemerides

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ABSTRACT

Context. Astronomical image metadata can be absent, altered, or less precise than the celestial information recorded in the image itself. Wide-field photographs often contain several planets whose changing relative positions provide an independent calendar and clock.

Aims. We test whether a single circular-fisheye photograph can be dated by combining a stellar astrometric solution with a blind search of planetary ephemerides.

Methods. Point sources were associated with Tycho-2 stars and fitted with the eight-parameter Barghini fisheye model. Unmatched bright detections were compared with topocentric positions of the planets from Mercury to Neptune on a daily grid from 1950 through 2026. After selecting the best day from two distinct planet–source associations, all image detections were reconsidered and the epoch was refined continuously with three planets.

Results. For a published image by F. Espenak, the blind daily search uniquely selected 2018 April 15. Mars, Jupiter, and Saturn were recovered independently, with measured–predicted separations of 0.064, 0.213, and 0.352 pixels at the stated time. The continuous three-planet minimum was 2018 April 15 07:08:16 UTC, 13 min 17 s after the published 06:55 UTC time. The stellar fit used 1213 associations and had 0.576-pixel RMS residual.

Conclusions. Planetary configurations can recover the date of a wide-field sky image and can provide a sub-day timing constraint. In this first example the 13-minute agreement is a best-fit offset, not a demonstrated timing precision: empirical astrometric perturbations move the minimum by several hours. A joint treatment of camera geometry, refraction, site, exposure interval, and ephemerides is the natural route to defensible uncertainties.

Key words. astrometry – methods: data analysis – techniques: image processing – planets and satellites: detection – atmospheric effects

1. Introduction

A sky photograph contains several kinds of time information even when its header does not. Stellar proper motions encode long intervals, sidereal rotation and star-trail lengths encode hours and exposure duration, and the Moon and planets move against the stellar background on timescales from hours to years. The planetary pattern is especially useful because it combines slow changes in relative configuration, which identify a date, with faster apparent motion and topocentric effects, which can constrain the time within that date.

Blind astrometric methods can identify the stellar coordinate system without reliable image metadata (Lang et al. 2010). Circular-fisheye imagery adds a strong, non-polynomial radial distortion and often reaches the horizon. Global all-sky camera models have therefore been developed for meteor and night-sky work (Borovička et al. 1995; Barghini et al. 2019; Jeanne et al. 2019). Once such a model maps every detection onto the celestial sphere, ephemeris matching becomes an astronomical dating problem rather than a visual identification exercise.

We demonstrate this idea on a public 180-degree image by Fred Espenak. Its web page supplies a date, time, site, camera, and lens (Espenak 2018), allowing an external test of the inferred epoch. These labels were used for evaluation and the final topocentric calculation; the published date and the named

planets were not supplied to the initial daily search. Section 2 describes the image, Sect. 3 gives the astrometric and ephemeris procedure, and Sect. 4 compares the recovered solution with the published metadata. We end with the attainable information and the remaining limitations.

2. Image and catalogue data

The test image is *Southern Milky Way and Planets*, photographed at Atacama Lodge near San Pedro de Atacama, Chile. The publisher states 2018 April 15 at 06:55 UTC, a Nikon D750, a Sigma 8-mm circular fisheye, and one 240-s exposure at $f/4.5$ and ISO 1600 (Espenak 2018). The published original is 4016×4016 pixels; we analysed the available 925×925 -pixel JPEG. The file metadata likewise report a Nikon D750, 8.0-mm focal length, 240 s, $f/4.5$, and ISO 1600. Its lens-model field instead gives 8.0 mm at $f/3.5$; we retain this discrepancy as metadata rather than using either lens label as a fit constraint.

We used a fixed local copy of the Tycho-2 catalogue (Høg et al. 2000), augmented by bright Hipparcos stars where required. Catalogue coordinates with two proper-motion components were propagated to the candidate epoch. A scan of the stellar residual against catalogue reference year was also made as a check on whether proper motion alone dated this recent image.

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Planets in Espenak's early-morning fisheye image
2018-04-15 06:55 UTC — Atacama Lodge, Chile

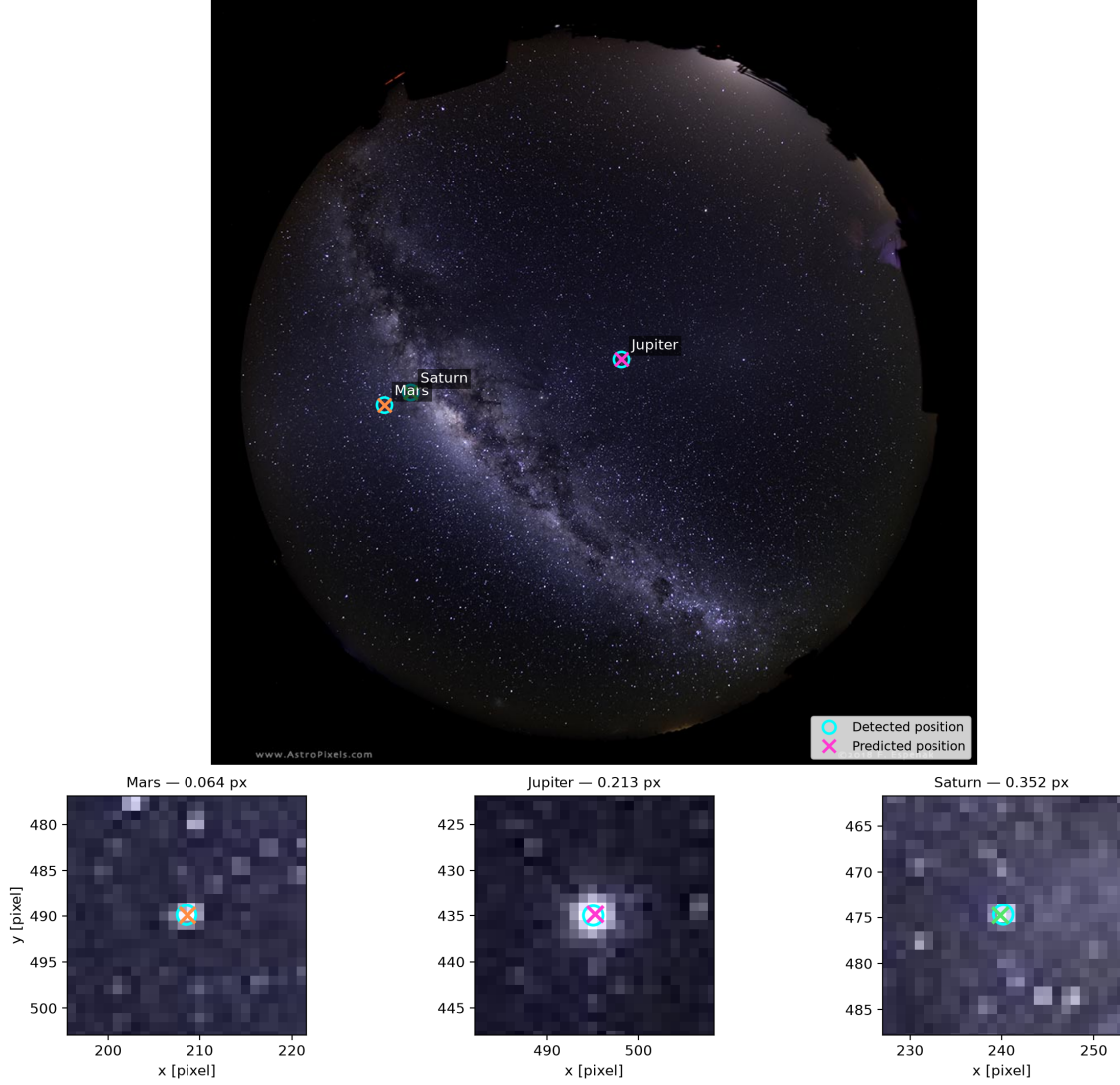


Fig. 1. Espenak's fisheye image with the three recovered planets. Cyan circles show the measured source centroids; coloured crosses show the topocentric ephemeris positions projected through the fitted camera model at the published epoch. The lower panels show the same coordinates at native pixel scale.

3. Method

3.1. Source association and global camera model

Point sources were detected after local background subtraction. Growing measurement windows retained broad and saturated sources. We obtained a small initial constellation with `tetra3`, ESA's open-source Python astrometric plate solver for star trackers (European Space Agency 2019). Its "lost-in-space" pattern matching identifies stars and determines the camera pointing in right ascension and declination without an initial pointing estimate. It therefore removes the need for a user-supplied initial sky pointing or WCS in our workflow, while its solution initialises the subsequent Barghini-model fit. This does not remove all prior information: `tetra3` requires an approximate field-of-view range and a compatible pattern database. Its distributed default database covers fields of view from 10° to 30° and stars down to magnitude 7 (European Space Agency 2019).

Progressive nearest-neighbour association then expanded the solution over the complete circular field. Every detected source

remained available for association. No catalogue identifier or saved solution was used to seed this demonstration.

The global projection is the eight-parameter model of Barghini et al. (2019), expressed with independent detector positions for the optical centre $O = (x_O, y_O)$ and reference direction $Z = (x_Z, y_Z)$. For a detector radius r_O about O , the projection angle u is

$$u(r_O) = Vr_O + S [\exp(Dr_O) - 1]. \quad (1)$$

The remaining parameter a_0 fixes the azimuthal orientation. The transformation from (u, b) about the optical axis to a celestial unit vector follows the spherical rotation in Eqs. (5), (6), and (11) of Barghini et al. (2019). We fit the eight parameters by minimising detector-plane residuals between measured centroids and catalogue directions projected into the image. In the present celestial solution Z is a fixed bootstrap reference direction; it should not be interpreted as a fitted terrestrial zenith.

The catalogue-to-camera fit is independent of the manufacturer's nominal fisheye projection. Its radial mapping, optical

centre, and orientation are therefore inferred from stars. Refraction was not included in this first planetary experiment. This omission is most consequential near the horizon and is carried into the empirical residuals rather than corrected cosmetically.

3.2. Blind date search and continuous refinement

The image positions of all detections were inverted through the fitted camera model to celestial unit vectors. We evaluated topocentric apparent positions of Mercury, Venus, Mars, Jupiter, Saturn, Uranus, and Neptune using Astropy’s solar-system transformations (Astropy Collaboration et al. 2022), whose built-in planetary model is based on the conventional analytical elements of Simon et al. (1994). The stated observing site was used for this topocentric step; geographical position was not fitted in the experiment reported here.

For every day from 1950 January 1 through 2026 December 31, we found the best assignment of two different planets to two different initially unmatched sources and recorded

$$Q_2(t) = \left[\frac{1}{2} \sum_{p=1}^2 \Delta\theta_p(t)^2 \right]^{1/2}, \quad (2)$$

where $\Delta\theta_p$ is the great-circle distance between a measured source and a planet. Restricting the blind stage to distinct source–planet pairs prevents a single bright object from satisfying both assignments. At the best day we again searched *all* detections, rather than preserving the unmatched classification, and accepted unique associations within 0.25° . This recovered a third planet. Holding those three associations fixed, we minimised the analogous three-planet RMS $Q_3(t)$ continuously within ± 2 d of the daily solution.

This staged procedure separates discovery from precision. The daily search asks whether the planetary configuration supplies a distinctive calendar date; the continuous fit measures where the astrometric residual is smallest on that date. To probe sensitivity to camera-model errors, we drew 2000 sets of three two-dimensional residual vectors from the fitted stellar residuals, added them to the planetary centroids, and repeated the fine timing search. This is an empirical perturbation test, not a formal confidence interval: it neglects correlated distortion, catalogue systematics, refraction, and uncertainty in the site.

4. Results

4.1. Stellar and lens solution

The image yielded 1255 detections, of which 1213 were associated with catalogue stars; 42 remained unmatched and none was withheld. The camera fit has an RMS residual of 0.576 px, a median of 0.367 px, and a 90th percentile of 0.745 px. The fitted optical centre is (463.79, 460.84) px. The radial coefficients are $V = 0.00310691$ rad px^{−1}, $S = 0.00710973$ rad, and $D = 0.00764440$ px^{−1}. The implied 90° radius is 441.17 px, and the central derivative of Eq. (1) is 0.18113° px^{−1}.

At the original 4016-pixel sampling, the fitted horizon radius would be about 1915 pixels under uniform rescaling, giving a 3830-pixel image circle. This is consistent with a circular fisheye nearly filling Espenak’s square frame. The agreement is a physical cross-check on the published Nikon D750 and 8-mm Sigma description; the fit itself used neither camera nor lens metadata. The left panel of Fig. 2 also shows that the fitted mapping is not exactly equidistant, illustrating why the catalogue-derived lens model is needed for the planetary comparison.

Table 1. Recovered planets at the published epoch.

Planet	Detection	Separation [pixel]
Mars	36	0.064
Jupiter	8	0.213
Saturn	54	0.352

4.2. Planetary date and time

The blind daily minimum occurred on 2018 April 15, the published date. Its Jupiter and Saturn assignments had a two-planet RMS of 0.0494° . After suppressing dates within 45 d of this solution, the next minimum was 1990 December 19 with 0.4002° RMS, eight times larger. Only 18 daily trials over the 77-year interval fell below 0.5° , so the global minimum is strongly distinguished.

Reconsidering all sources on the selected date independently recovered Mars, Jupiter, and Saturn as detections 36, 8, and 54. At the stated time their detector-plane residuals were 0.064, 0.213, and 0.352 pixels, respectively (Fig. 1 and Table 1). The other searched planets were below the local horizon. The fixed-association continuous minimum was 2018 April 15 07:08:16 UTC with $Q_3 = 0.0460^\circ$, an offset of 13 min 17 s from the stated 06:55 UTC.

The small offset is an accuracy check against external metadata, rather than a 13-minute uncertainty. The interval over which $Q_3 < 0.1^\circ$ is about 14 h. In the empirical residual-vector experiment, the 16th, median, and 84th percentiles of the fitted-time displacement were -2.92 , -0.08 , and $+3.00$ h. The 240-s exposure introduces a further two-minute convention shift if the published time refers to an endpoint rather than mid-exposure. Proper motion did not usefully constrain this recent date: among the 1127 matched stars with both components, a reference-year scan changed the RMS by only 9.8×10^{-6} px between 2009 and 2018.

5. Perspectives

Dating a single sky image to the correct day is useful wherever provenance is incomplete: digitised plate and film collections, meteor and auroral archives, environmental all-sky cameras, and historical or public astrophotography. Large photographic archives already open century-long time baselines (Grindlay et al. 2009); an astronomical timestamp can expose catalogue mistakes, order undated sequences, and connect an image to observing logs or transient events. In particular, a date recovered for a historical image containing a nova could place that observation on the nova light curve and constrain its rise, peak, or decline. This is valuable for epochs before systematic patrol cameras supplied dense time-series coverage. A planetary solution is also independent of file timestamps and many camera-clock failures.

The 13-minute agreement in this example is interesting because it comes from the celestial scene in one static exposure, but it should be understood as a target for a coupled model. Planetary angular velocities near the observed configuration convert small astrometric biases into hours of timing shift. A fit that simultaneously includes a physical zenith, latitude, atmospheric refraction and extinction, camera geometry, and the finite exposure should separate these effects. Star-trail angular length can supply exposure duration; the distribution of photometric residual against airmass can constrain the zenith and extinction; and the resulting terrestrial orientation can supply local sidereal time.

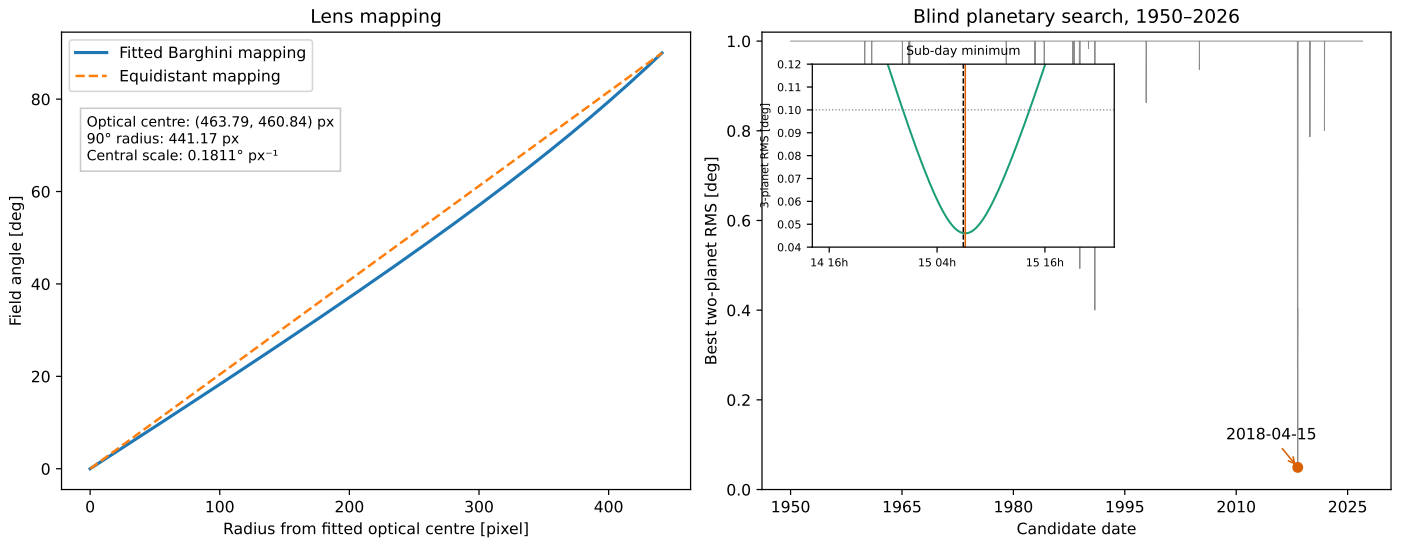


Fig. 2. Left: fitted Barghini radial mapping and an equidistant mapping with the same 90° radius. Right: the blind two-planet objective over 1950–2026, clipped at 1° for display. The inset gives the continuous three-planet objective around the selected date; dashed and orange vertical lines mark the published and best-fitting epochs, respectively.

Together these observables could turn the sharp formal planetary minimum into a defensible sub-hour uncertainty and might constrain an unknown observing latitude as well as the epoch. Testing that prospect requires multiple images, deliberately blinded metadata, and injected perturbations.

6. Conclusions

A global stellar astrometric solution and a blind planetary ephemeris search recovered the published date of a 180-degree fisheye photograph over a 1950–2026 search interval. Three independently recovered planets placed the continuous minimum 13 min 17 s from the stated time, while their predicted positions agreed with the measured centroids to better than 0.36 pixel. The catalogue-fitted Barghini mapping was consistent with the published circular 8-mm fisheye geometry and avoided assuming an ideal manufacturer projection.

The experiment establishes calendar recovery and close agreement with an external timestamp. Its present several-hour empirical sensitivity shows where the scientific work now lies: coupled refraction, orientation, site, exposure, and camera modelling, followed by validation on a blinded image set.

Acknowledgements. We thank Fred Espenak for publishing the image and its observing metadata. The image remains credited to and copyrighted by its photographer; its use here is for scientific analysis. This work made use of Astropy and tetra3.

Data and code availability

The figure-generation code, numerical summary, and verification tests accompany this manuscript repository. The complete analysis code, fitted parameters, source associations, and machine-readable planetary matches will be deposited as a versioned Zenodo record upon acceptance; the final DOI will replace this statement. The input photograph is available from the publisher’s linked image page, subject to the photographer’s terms (Esenak 2018).

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